

Control Algorithm Parameters for Patchouli Dryer

Below is a summary table of the key control parameters and rules for the ESP32-based patchouli dryer. Each value is chosen based on the literature and drying physics, with references provided for justification.

Parameter	Specification / Rule	Rationale / References
Operating Temperature (air)	Target 45 °C; Safety cutoff ~60 °C	45 °C is chosen because forced drying at 45 °C gave the best balance of fast moisture removal and oil yield in patchouli [28†L212-L218] . Temperatures >60 °C risk volatile oil loss (ASTRA dryer reached 95 °C and noted oil loss [3†L61-L64]), so the heater is limited to ~60 °C max for safety.
Relative Humidity (RH)	Target $\leq 60\%$ (run dehumidifier if RH > 60 %)	Lower ambient RH speeds drying. In one study, patchouli dried at ~33 °C/70 %RH took ~45 h to reach ~14% moisture [28†L177-L184] . Keeping RH <60% (via Peltier/fan) prevents saturation of the air and mold risk.
Sampling & Filtering	Load cell: sample ~10 Hz, apply moving-average (e.g. last 5 readings). SHT31: sample ~1 Hz (no heavy filtering needed).	Raw weight readings are noisy. Use a low-pass or moving-average filter on the HX711 data to smooth spikes [9†L23-L32] . SHT31 temperature/humidity is stable at ~1s intervals.
ΔWeight Threshold & Drying Rate	ΔWeight ≈ 0 (e.g. <0.1 % of batch per interval); Drying rate ≈ 0	Drying is complete when mass stops changing [5†L144-L148] . A small deadband (~0.1% of total, ≈ 5 g for a 5 kg batch) accounts for sensor noise [9†L23-L32] . Both $\Delta W \approx 0$ and near-zero drying rate indicate equilibrium.
Stop Condition	End drying when ΔWeight ≈ 0 over consecutive checks (e.g. 2+ readings). Then Heater=OFF , Fan=ON (cooling for ~5 min) , then FINISH .	In practice, once weight is stable for back-to-back readings [5†L144-L148] , the heater is turned off. The fan continues (cooling mode) briefly to purge heat, then the process ends. This matches literature advice to stop at constant weight [5†L144-L148] .

Control Logic: Based on the above parameters, the ESP32 runs a rule-based loop. During drying, the heater is turned on whenever chamber temperature <45 °C, and off if ≥ 45 °C or safety limit reached. If RH >60 %, the Peltier dehumidifier is enabled until RH falls. The load-cell weight is monitored continuously (with filtering) to compute Δ Weight (mass change) and drying rate. When Δ Weight is effectively zero (within

threshold) for two successive checks 【5†L144-L148】 , the controller concludes the leaves are fully dry. At that point the heater is turned off and the fan runs for a short cooling period before stopping the cycle.

This multi-criteria rule-based approach (temperature, humidity, and mass-change) is grounded in drying physics: 45 °C forced-air drying was shown to yield ~2.6 % oil with 4 h drying 【28†L212-L218】 , while shade (high RH) required ~45 h 【28†L177-L184】 . Thus 45 °C is a practical setpoint. Likewise, stopping when weight stabilizes (rather than a fixed time) ensures consistent dryness, as recommended by past studies 【5†L144-L148】 . The parameters above therefore reflect best practices from the literature on patchouli and herb drying.

Sources: Peer-reviewed studies of patchouli drying and drying control systems 【28†L212-L218】 【5†L144-L148】 【3†L61-L64】 【9†L23-L32】 were used to set these parameters and logic. They confirm that 40–45 °C is ideal for patchouli (45 °C chosen here) and that weight stabilization is a reliable stop criterion 【28†L212-L218】 【5†L144-L148】 , while noting that high heat (≥ 95 °C) can degrade oil 【3†L61-L64】 , so a ~60 °C safety cutoff is prudent. The weight sampling/filtering strategy is supported by practices for load-cell sensors 【9†L23-L32】 .
